AIML231 — Techniques in Machine Learning

Week 1

Machine Learning Tasks

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Overview

★ AI and machine learning

★ Machine learning scope: data, task, model, and algorithm

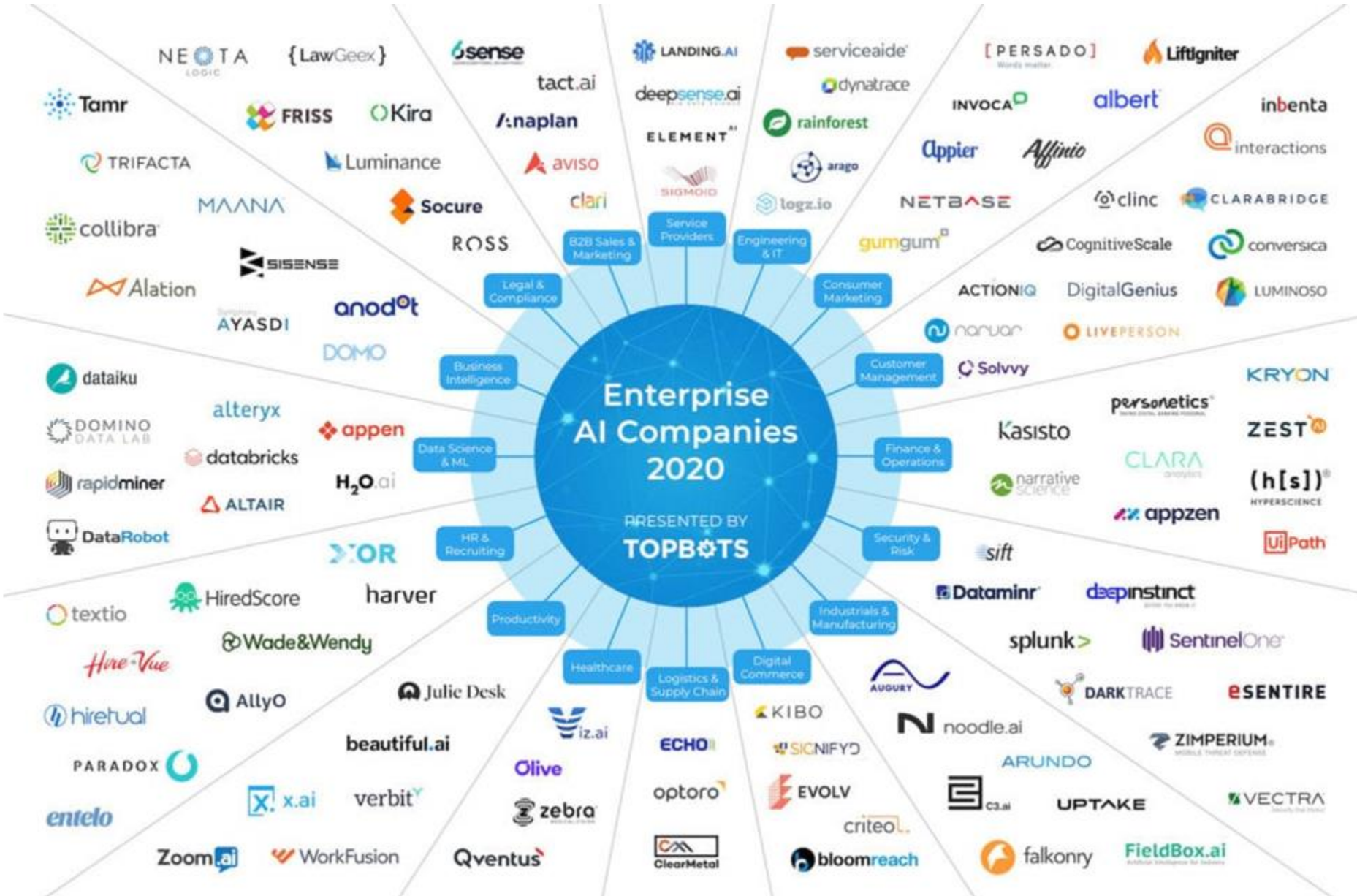
★ Data handle by machine learning

★ Machine learning tasks

★ Python Programming for ML Basics

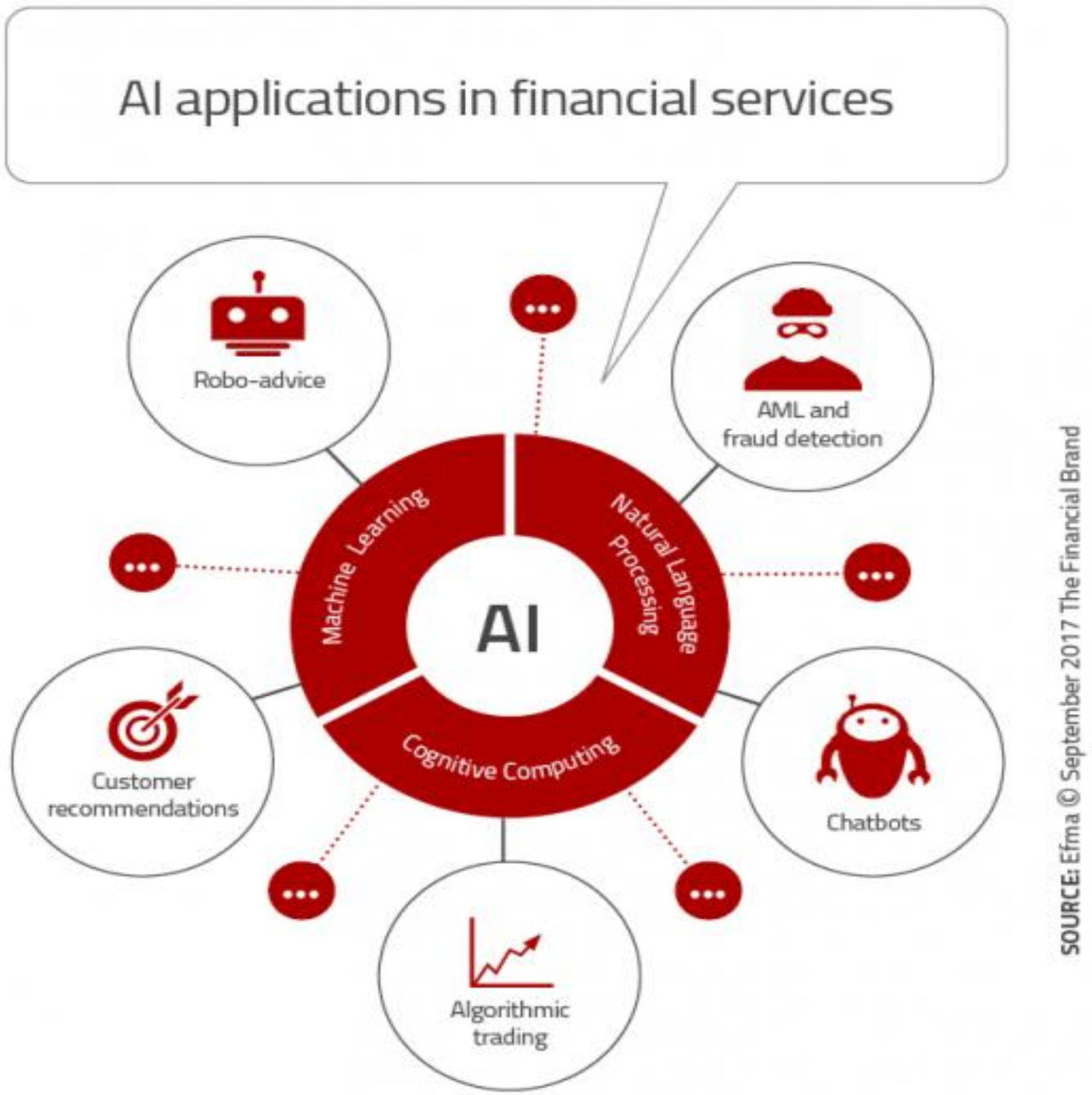
Artificial Intelligence Companies

Artificial Intelligence Companies



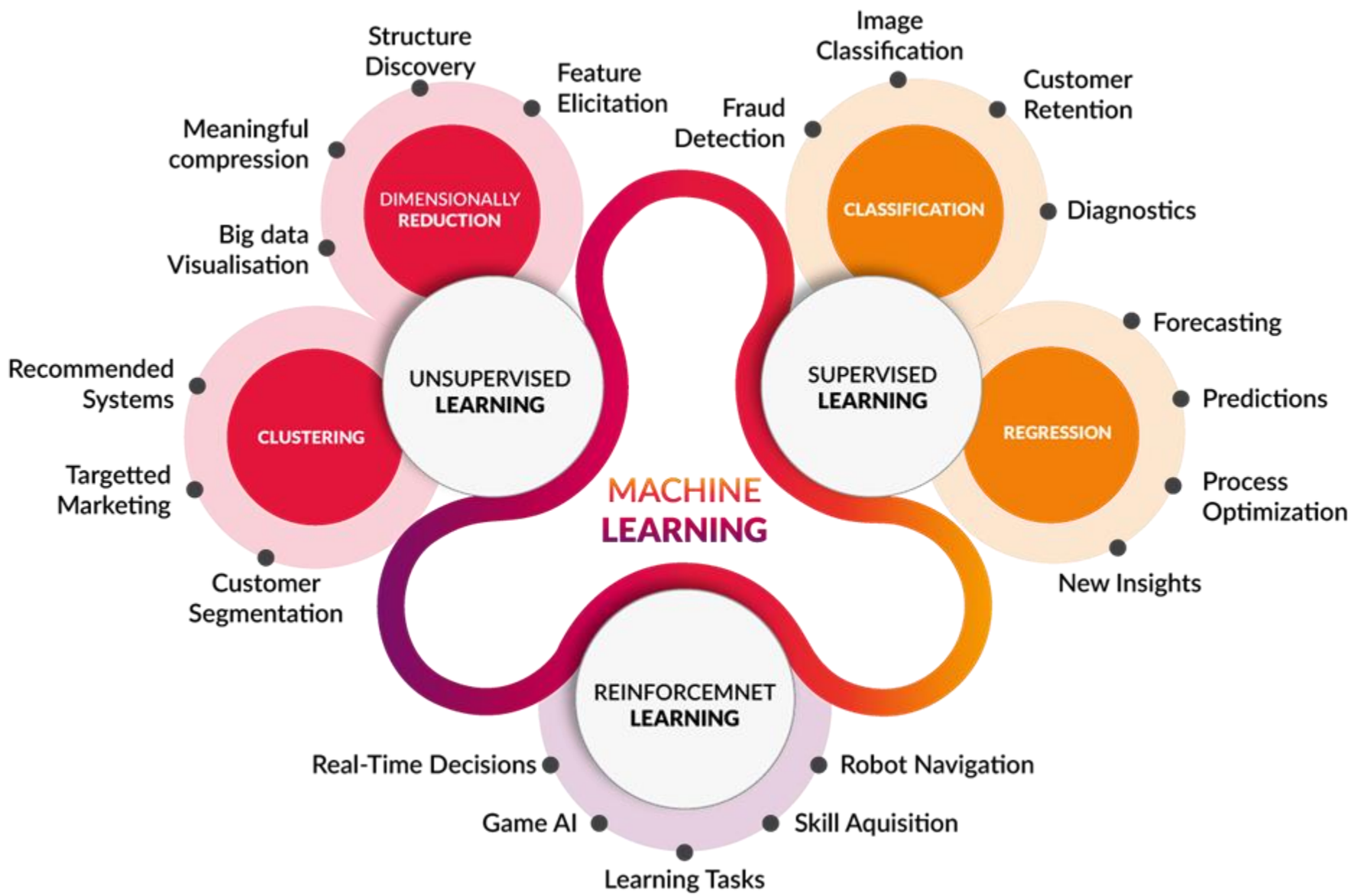
cf in Aotearoa :
https://newzealand.ai/nz-ai-companies

Not This - This is Application



Not this either - this is the process of ML

ML Tasks - 3-way split



http://www.cognub.com/index.php/cognitive-platform/

Supervision available to the learner

lots:

Supervised learning: The environment contains a teacher that provides the correct response for certain environmental states. The goal is for the learner to output the correct response: “do what the teacher would do”.

none:

Unsupervised learning: No such teacher to say whether the learner’s output is correct. Instead, the learning system has an internally defined teacher with a prescribed goal that does not need utility feedback of any kind.

some:

Reinforcement learning: Again, no such teacher to say whether the learner’s output is correct. Instead of a label, the environment provides reward or punishment to indicate the utility of actions that were actually taken by the system.

The main divisions within learners

	Supervised Learning	Unsupervised Learning
Discrete	Classification (categorisation)	Clustering
Continuous	Regression	Dimension Reduction

(leaving Reinforcement learning aside)

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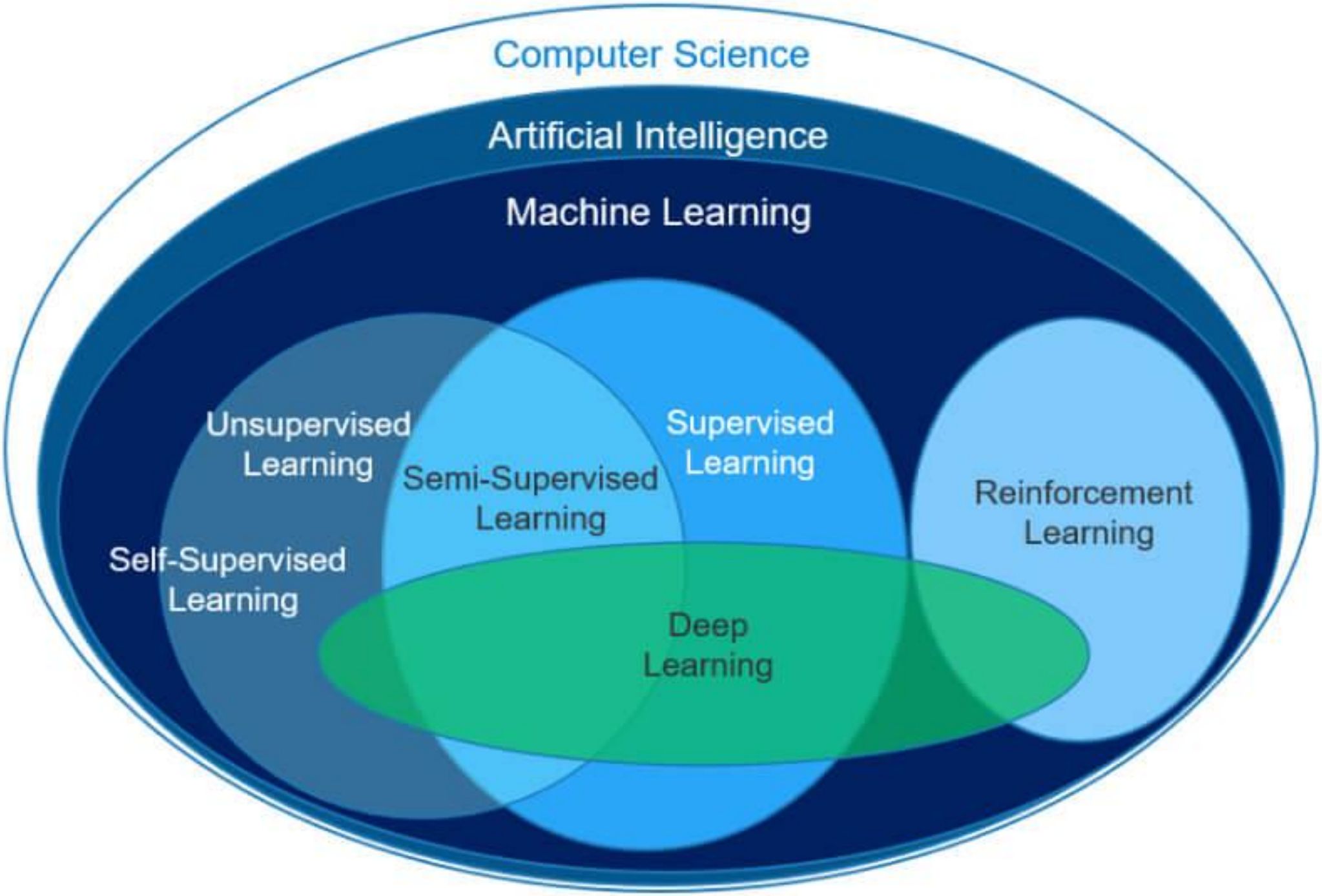
Semi-supervised Learning?

Self-Supervised Learning?

Adapted from: https://www.slideshare.net/GirishKhanzode/supervised-learning-52218215

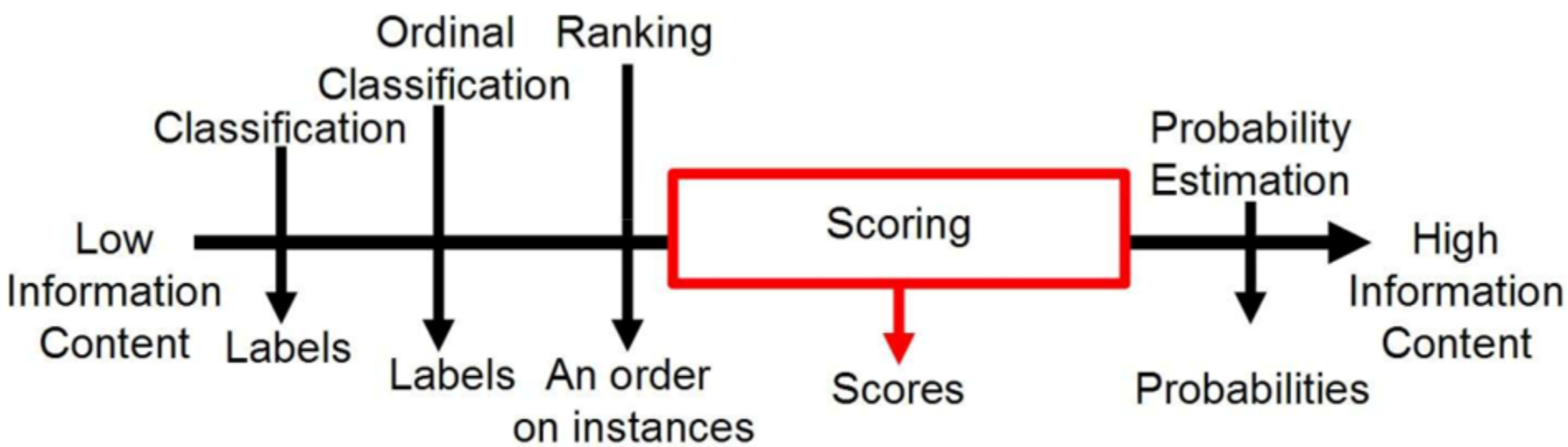
More ML Tasks



https://medium.datadriveninvestor.com/machine-learning-in-10-minutes-354d83e5922e

Outcomes (outputs) – can be simple… or more informative

- Notice that different outcomes can have different amounts of information content
- *e.g. here, a classifier classifies☺ some novel input pattern*



notice we can think of the classifier's response a "prediction", of a kind

Prediction Outcomes

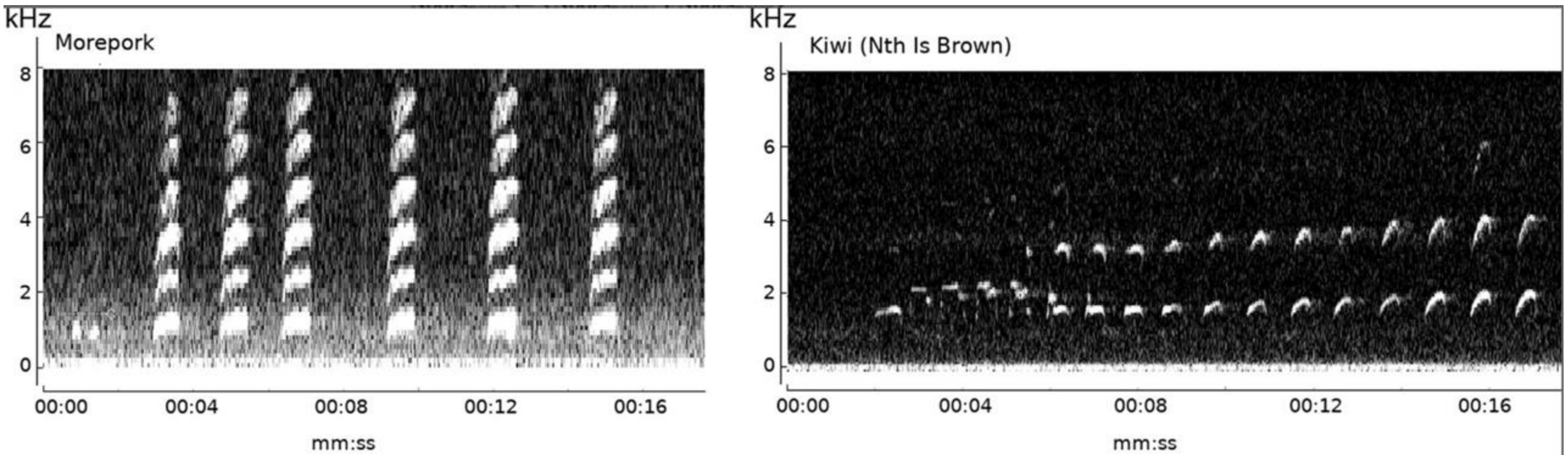
https://www.researchgate.net/publication/220699539_Smooth_Receiver_Operating_Characteristics_smROC_Curves

Types of Input

- i.i.d. (**i**ndependent, **i**dentically **d**istributed)

V1	V2	V3	V4	Class
3.6216	8.6661	-2.8073	-0.44699	1
4.5459	8.1674	-2.4586	-1.4621	1
3.866	-2.6383	1.9242	0.10645	1
3.4566	9.5228	-4.0112	-3.5944	1
0.32924	-4.4552	4.5718	-0.9888	1
4.3684	9.6718	-3.9606	-3.1625	1
3.5912	3.0129	0.72888	0.56421	1
2.0922	-6.81	8.4636	-0.60216	1
3.2032	5.7588	-0.75345	-0.61251	1
1.5356	9.1772	-2.2718	-0.73535	1
1.2247	8.7779	-2.2135	-0.80647	1

- or a sequence, like audio, video, time series …



https://www.avianz.net

- or image, graphs, multi-modal data?

Classification

- Classification: a type of supervised machine learning task where the goal is to assign an input to one of several predefined categories or classes

V1	V2	V3	V4	Class
3.6216	8.6661	-2.8073	-0.44699	1
4.5459	8.1674	-2.4586	-1.4621	1
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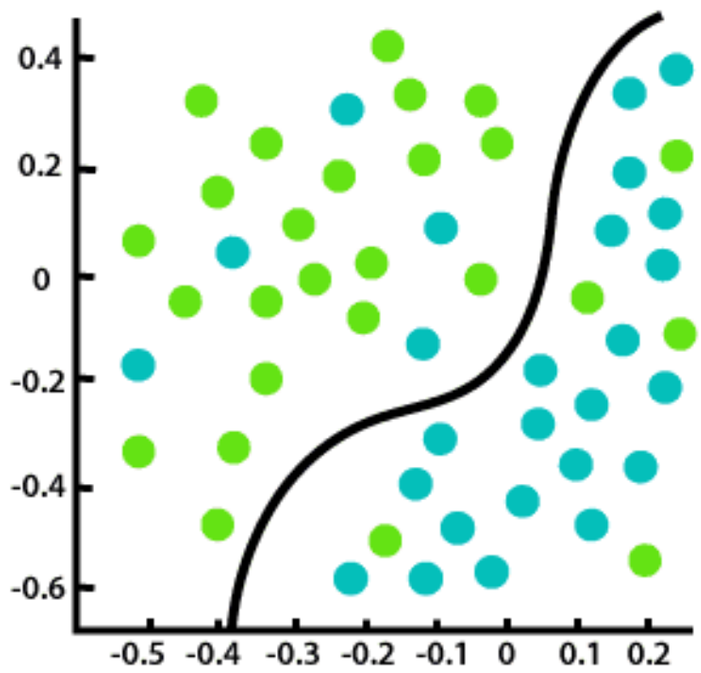
- Binary Classification: two classes
- Multiclass Classification: more than two classes
- Multilabel Classifications: each instance can belong to multiple classes

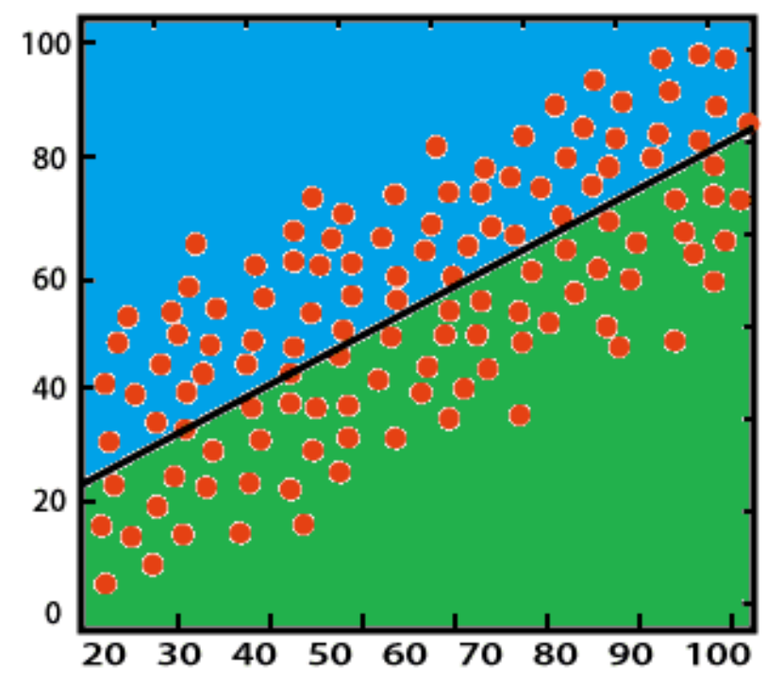
	Sepal length	Sepal width	Petal length	Petal width	Type
1	5.1	3.5	1.4	0.2	Iris setosa
2	4.9	3.0	1.4	0.2	Iris setosa
...					
51	7.0	3.2	4.7	1.4	Iris versicolor
52	6.4	3.2	4.5	1.5	Iris versicolor
...					
101	6.3	3.3	6.0	2.5	Iris virginica
102	5.8	2.7	5.1	1.9	Iris virginica
...					

Regression

- Regression: a type of supervised machine learning task where the goal is to predict a continuous numerical value based on input features
- find the relationship between the dependent variable (target) and one or more independent variables (features)



Classification



Regression

Clustering

- Clustering: finding groups of items that are “similar”
- *Unsupervised,* class of an example is not known
- Success often measured subjectively – it is <u>fundamentally ill-posed</u>!

	Sepal length	Sepal width	Petal length	Petal width	Type
1	5.1	3.5	1.4	0.2	Iris setosa
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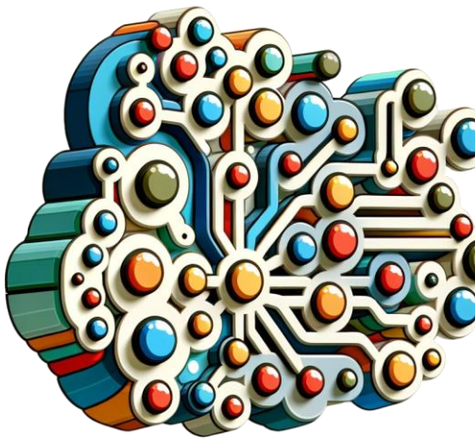
…					

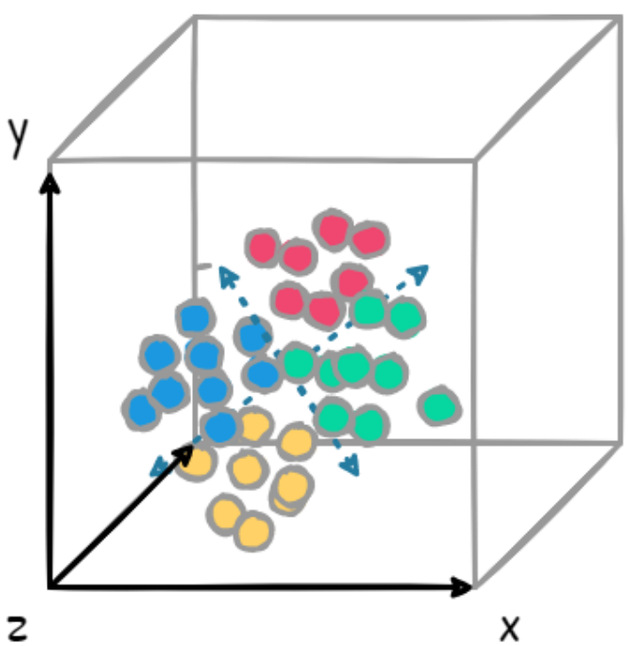


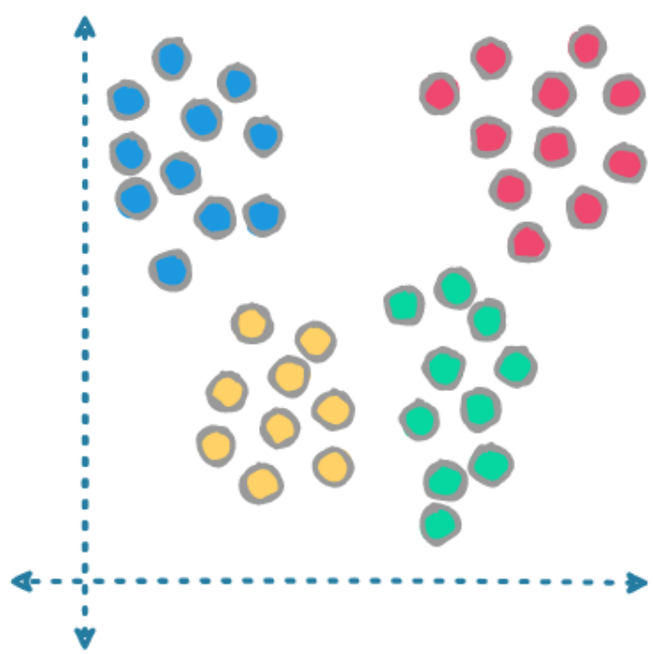
Dimensionality Reduction

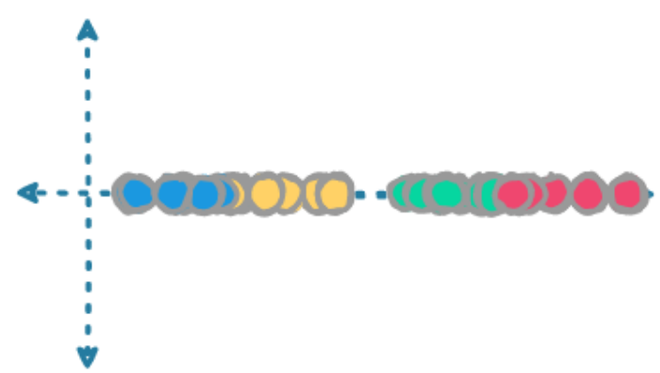
Dimensionality reduction: reduce the number of features (variables) in a dataset while retaining as much relevant information as possible, e.g. Principle Components Analysis (PCA), and others

Raw data is often high-dimensional, which is a problem:

1. data with >3 dimensions is hard for humans to interpret / work with!
2. ML algorithms also struggle with high-dimensional data (ultimately, due to the curse of dimensionality)







Some Others

Tasks	Description	Algorithms	Examples
Classification	Predict if a data point belongs to one of the predefined classes. The prediction will be based on learning from a known data set.	Decision trees, neural networks, Bayesian models, induction rules, k-nearest neighbors	Assigning voters into known buckets by political parties, e.g., soccer moms Bucketing new customers into one of the known customer groups
Regression	Predict the numeric target label of a data point. The prediction will be based on learning from a known data set.	Linear regression, logistic regression	Predicting unemployment rate for next year Estimating insurance premium
Anomaly detection	Predict if a data point is an outlier compared to other data points in the data set.	Distance based, density based, local outlier factor (LOF)	Fraud transaction detection in credit cards Network intrusion detection
Time series	Predict the value of the target variable for a future time frame based on historical values.	Exponential smoothing, autoregressive integrated moving average (ARIMA), regression	Sales forecasting, production forecasting, virtually any growth phenomenon that needs to be extrapolated
Clustering	Identify natural clusters within the data set based on inherit properties within the data set.	k-means, density-based clustering (e.g., density-based spatial clustering of applications with noise [DBSCAN])	Finding customer segments in a company based on transaction, web, and customer call data
Association analysis	Identify relationships within an item set based on transaction data.	Frequent Pattern Growth (FP-Growth) algorithm, Apriori algorithm	Find cross-selling opportunities for a retailer based on transaction purchase history

https://tanthiamhuat.wordpress.com/data-mining/

Python Programming Basics for ML

Why Python:

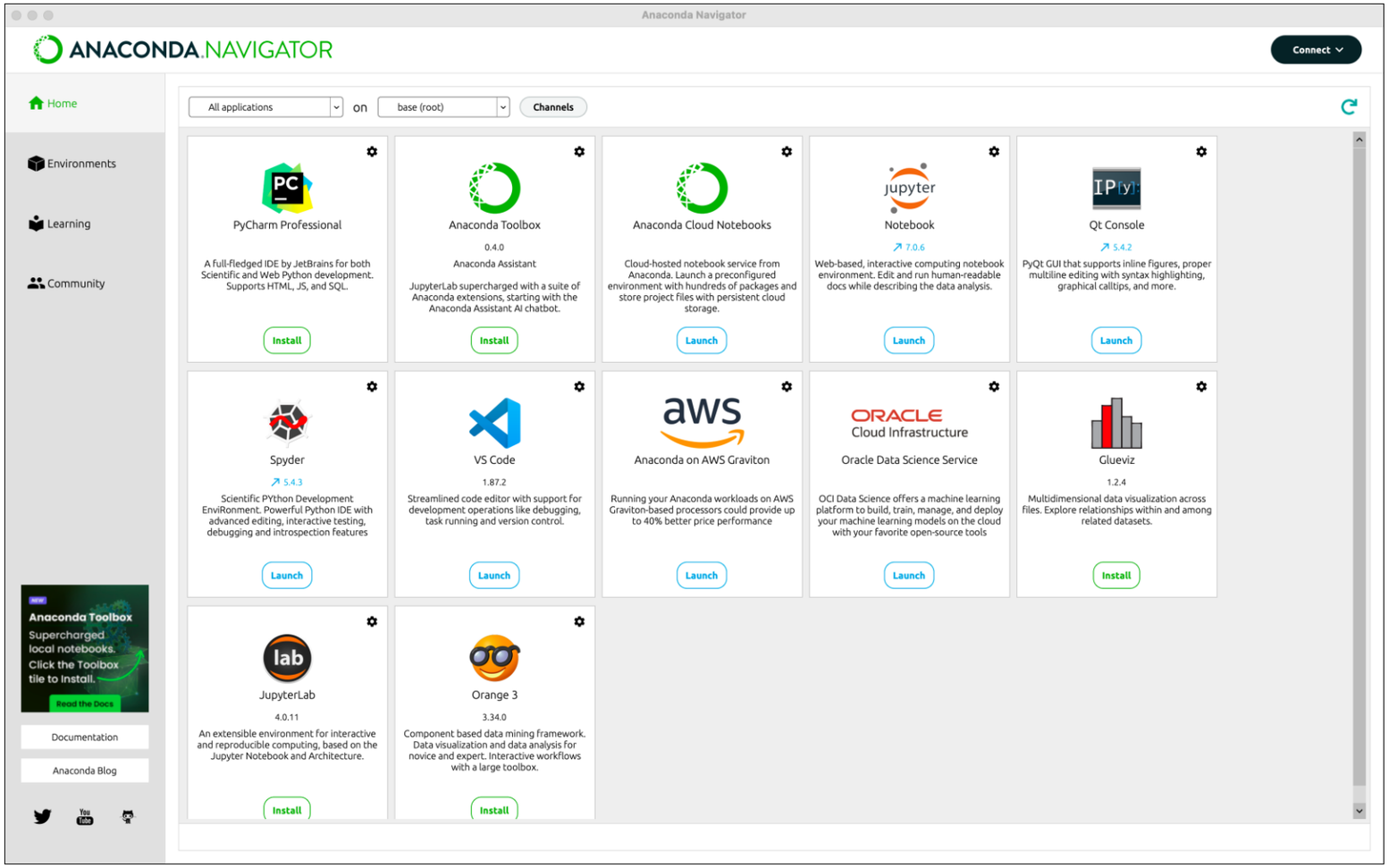
- Easy to learn: Simple syntax, readable code
- Rich ecosystem: Libraries like NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch
- Great community support: Extensive documentation and active forums
- Integration capabilities: Works with databases, web applications, and big data platforms.

Setting up Environment

- Option 1: Download from Official Website
  https://www.python.org/downloads/

- Option 2: Install via Anaconda (Recommended):
  https://www.anaconda.com/download

  - Anaconda provides Python + Jupyter Notebook + ML libraries pre-installed



Python Modules

- Python modules and libraries are essential for organizing and reusing code efficiently

- A module is a single Python file (.py) that contains reusable code such as functions, classes, and variables

  - E.g. create a Python file "math_operations.py" and define some functions
  - In another Python file, import "math_operations" to use its functions

```python
# math_operations.py

def add(a, b):
    """Returns the sum of two numbers."""
    return a + b

def multiply(a, b):
    """Returns the product of two numbers."""
    return a * b

def subtract(a, b):
    """Returns the difference of two numbers."""
    return a - b
```

```python
# main.py

# Import the module
import math_operations

# Use functions from the module
num1 = 10
num2 = 5

sum_result = math_operations.add(num1, num2)
product_result = math_operations.multiply(num1, num2)
difference_result = math_operations.subtract(num1, num2)

print(f"Sum: {sum_result}")         # Output: Sum: 15
print(f"Product: {product_result}") # Output: Product: 50
print(f"Difference: {difference_result}") # Output: Difference: 5
```

Use Modules

- Import the whole module: *import math_operations*
- import only specific functions:

```
from math_operations import add, multiply

print(add(3, 7))      # Output: 10
print(multiply(4, 6)) # Output: 24
```

- Import with an Alias

```
import math_operations as mo

print(mo.add(2, 8))  # Output: 10
```

Python Libraries

- A library is a collection of modules that provides pre-written functions for specific tasks

- Unlike a module, a library consists of many Python files (modules) grouped together

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5])
print(arr.mean())  # Output: 3.0
```

Key libraries:

- NumPy → Efficient array computations
- Pandas → Data manipulation
- Matplotlib → Data visualization
- Scikit-learn → ML models

Standard Python Modules vs. External Libraries

- Standard Modules: built-in, no installation needed, e.g., *math, os, sys, random, datetime*

- External Libraries: require installation via ***pip*** or ***conda,*** *e.g., numpy, pandas, scikit-learn, matplotlib*

  - pip install numpy
  - conda install numpy

  - *Use pip in a standard Python environment*
  - *Use Conda with Anaconda, more effective to manage package dependencies*

Example: ML with Scikit-learn

Steps in a Machine Learning Model

- Load data
- Preprocess data
- Train a model
- Evaluate model
- Make predictions

```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# Sample dataset
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_s

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, y_pred)
print(f"Mean Squared Error: {mse}")
```